

INCIDENT INVESTIGATION REPORT

Unauthorized Bluetooth Device Connection — Enterprise Workstation
IR Case Reference: IR-001

Classification: UNCLASSIFIED — Sanitized for Portfolio Use

Report Date	May 2026
Examined By	Oliver S. Degraw — Cybersecurity Specialist / ISSO
Date of Examination	27–29 May 2026
System Examined	NIPR-WKS-001 (sanitized hostname)
Operating System	Windows 11 Enterprise (Build 26100)
Domain	[Enterprise Domain — sanitized]
Case Status	Closed — User Identified; Systemic Remediation In Progress

1. Executive Summary

A forensic investigation was conducted on an enterprise Windows 11 workstation following the discovery of an active Bluetooth capability in violation of applicable cybersecurity policy. Registry examination confirmed an unauthorized Bluetooth audio device had connected to the workstation outside normal duty hours. The investigation identified both an individual policy violation and a systemic security control gap affecting the broader workstation population at the installation.

This report documents forensic findings, artifacts collected, pending investigative actions, and formal remediation recommendations across immediate, short-term, and long-term timeframes.

2. Background

During a routine security review, Bluetooth functionality was found active on a workstation where Bluetooth is required to be disabled per applicable DoD and Air Force cybersecurity policy. A forensic examination was initiated to determine the scope, nature, and root cause of the violation.

The investigation revealed two distinct findings: (1) an individual unauthorized device connection incident, and (2) a systemic procedural gap in the current workstation imaging process that leaves Bluetooth controls unverified and unenforced at the enterprise level.

3. Individual Incident Findings

3.1 Unauthorized Bluetooth Device Confirmed

Registry examination confirmed an unauthorized Bluetooth audio device (consumer-grade wireless headphones) was paired with the workstation. The device record was located in the Windows registry Bluetooth port parameters hive and contained the following:

Device Type	Consumer Bluetooth Audio Device (manufacturer sanitized)
Bluetooth MAC Address	XX:XX:XX:XX:XX:XX (sanitized)
Last Connected	March 2026 — 0401 local time
Connection Context	Outside normal duty hours
Registry Location	HKLM\SYSTEM\CurrentControlSet\Services\BTHPORT\Parameters\Devices\[MAC]

3.2 Bluetooth Hardware Active — BIOS Control Step Not Completed

The established imaging procedure requires a technician to manually disable Bluetooth in the system BIOS following each reimage. Examination confirmed Bluetooth hardware was fully operational at time of examination, indicating the post-image BIOS disable step was not completed on at least two separate occasions:

- Prior to March 2026 — Bluetooth was active, allowing the unauthorized device to connect
- May 2026 reimage — Bluetooth hardware was not disabled post-reimage, leaving the workstation non-compliant after servicing

3.3 Bluetooth Service Not Disabled at Windows Level

The Bluetooth Support Service was found configured as Manual start rather than Disabled. No Group Policy Object (GPO) exists in the applied policy set to enforce the Disabled state at the Windows service level.

Registry Key	HKLM\SYSTEM\CurrentControlSet\Services\bthserv
Start Value Found	0x00000003 (3) — Manual
Expected Value	0x00000004 (4) — Disabled
GPO Enforcement	None — no GPO configures this setting
Control Method	Manual BIOS disable by imaging technician only

3.4 Secure Boot Confirmed Active

Secure Boot was confirmed enabled on the workstation. This eliminates the possibility that a standard user account could have accessed BIOS settings to enable Bluetooth, as BIOS access is password-protected. Any BIOS-level change would require administrator-level access or knowledge of the BIOS password.

3.5 Reimage Confirmed — May 2026

Forensic examination confirmed the workstation was reimaged in May 2026. Three independent sources corroborated this finding:

- setupapi.dev.log — Multiple sequential Boot Session entries consistent with a fresh OS installation
- Kernel-PnP Event ID 440 — Device settings flagged as migrated from previous OS installation, including the unauthorized device record
- OS Install Date — Windows reports install date consistent with reimage; no upgrade migration log exists, confirming clean reimage rather than in-place upgrade

3.6 User Identification — Domain Controller Coordination

Local Security event log retention did not extend back to the date of the original unauthorized connection, which is consistent with normal log retention behavior and does not indicate deliberate tampering. The investigation was escalated to the network operations team, who coordinated a Domain Controller log pull. Security Event IDs 4624 and 4672 were reviewed for the incident time window, resulting in successful identification of the user account active at the time of the Bluetooth connection.

Further log analysis was determined unnecessary following identification of the systemic registry-level control gap documented in Section 4. The root cause — absence of GPO enforcement for the Bluetooth service start value — was established as the primary vulnerability driving both the individual incident and the broader installation-wide risk.

4. Systemic Finding — Bluetooth Security Control Gap

⚠ SYSTEMIC FINDING: This control gap extends beyond the individual incident and affects the broader NIPR workstation population at the installation.

Investigation revealed that the Bluetooth disable control for NIPR workstations relies entirely on a manual BIOS-level step performed by a technician during the imaging process. This control is not enforced by Group Policy, not incorporated into the baseline image, and not subject to automated verification or audit. Specifically:

- No GPO exists in the applied policy set that disables the Bluetooth Support Service or restricts Bluetooth hardware
- The Windows registry bthserv Start value is not set to Disabled as part of the imaging process — it defaults to Manual on all imaged machines
- The sole security control is a manual technician action in BIOS post-image, which is subject to human error and produces no auditable record
- This gap likely exists across multiple NIPR workstations imaged using the same procedure

Risk Assessment:

A single missed step by any imaging technician results in a non-compliant workstation with active Bluetooth capability. This creates an exploitable attack surface in violation of DoD and Air Force cybersecurity requirements. The current control provides no defense-in-depth and no compensating control if the BIOS step is skipped.

5. Recommended Corrective Actions

5.1 Immediate

- Disable Bluetooth in BIOS on affected workstation immediately to return it to a compliant state
- Audit all NIPR workstations currently in service for Bluetooth hardware status using Get-PnpDevice PowerShell cmdlet and bthserv Start value — identify all non-compliant machines

5.2 Short Term

- Create and deploy a GPO to set the bthserv service Start value to Disabled (0x4) across all NIPR workstations — this removes reliance on the manual BIOS step as the sole control and provides GPO-enforced, auditable compliance
- Update the post-imaging checklist to include a mandatory Bluetooth BIOS disable verification step with technician sign-off
- Implement a second-technician verification requirement for the BIOS disable step on all NIPR reimages

5.3 Long Term

- Incorporate Bluetooth disable into the baseline NIPR image configuration so the control is applied automatically rather than manually
- Implement periodic automated compliance scanning for bthserv Start value across all domain-joined NIPR workstations using PowerShell or endpoint management tooling
- Document this finding as a formal configuration management deviation and submit through the Risk Management Framework process for tracking and remediation verification

7. MITRE ATT&CK Mapping

Technique ID	Technique Name	Relevance
T1200	Hardware Additions	Unauthorized Bluetooth peripheral connected to enterprise workstation
T1052	Exfiltration Over Physical Medium	Bluetooth connection creates potential data exfiltration channel
T1562.001	Impair Defenses: Disable or Modify Tools	Absence of GPO enforcement represents a defensive control gap exploitable by insiders

7. Applicable Policy

- DoD Instruction 8500.01 — Cybersecurity
- DAFMAN 17-1301 — Computer Security (COMPUSEC)
- DISA STIG: Windows 11 — WN11-CC-000035 (Bluetooth must be disabled)
- Privileged User Agreement (PUA) — Administrator responsibilities for maintaining system compliance

8. Evidence Preserved

The following artifacts were collected during examination:

Artifact	Description
bt_devices.reg	Registry export of BTHPORT\Parameters\Devices confirming unauthorized device pairing record and timestamp
bthserv.reg	Registry export of Bluetooth service key confirming non-compliant Manual start value
setupapi_log.txt	setupapi.dev.log entries confirming reimage Boot Sessions and installation date
Screenshot_01.png	PowerShell FILETIME conversion confirming unauthorized connection timestamp
Screenshot_02.png	Registry Editor view of unauthorized device key under BTHPORT
Screenshot_03.png	Registry query output confirming Bluetooth service Start value of Manual (non-compliant)
Screenshot_04.png	User profile timestamps consistent with reimage date
Screenshot_05.png	Kernel-PnP Event ID 440 — device migration entries confirming device record survived reimage
Screenshot_06.png	Kernel-PnP Event ID 400/410 — device configured and started entries
Screenshot_07.png	Kernel-PnP Event ID 420 — device deletion entry confirming remediation

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